Logarithmic Functions & Properties

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Converting between exponential and logarithmic form**

Rewrite $2^5 = 32$ in logarithmic form, and $\log_3 81 = 4$ in exponential form.

$2^5 = 32 \Leftrightarrow \log_2 32 = 5$. $\log_3 81 = 4 \Leftrightarrow 3^4 = 81$.

2 **[Foundational] • Target: Converting between exponential and logarithmic form**

Evaluate $\log_5 125$ and $\log_2 \frac{1}{8}$ without a calculator.

$\log_5 125 = 3$ since $5^3 = 125$. $\log_2 \frac{1}{8} = -3$ since $2^{-3} = \frac{1}{8}$.

3 **[Foundational] • Target: Properties of logarithms**

Use logarithm properties to expand $\log\left(\frac{x^3\sqrt{y}}{z^2}\right)$ fully.

$\log\left(\frac{x^3\sqrt{y}}{z^2}\right) = \log(x^3) + \log(y^{1/2}) - \log(z^2) = 3\log x + \frac{1}{2}\log y - 2\log z$.

4 **[Foundational] • Target: Properties of logarithms**

Combine $2\log x - \frac{1}{3}\log y + \log z$ into a single logarithm.

$2\log x - \frac{1}{3}\log y + \log z = \log(x^2) - \log(y^{1/3}) + \log z = \log\left(\frac{x^2 z}{y^{1/3}}\right)$.

5 **[Foundational] • Target: Properties of logarithms**

Use the change of base formula to evaluate $\log_7 50$, to three decimal places.

$\log_7 50 = \frac{\ln 50}{\ln 7} = \frac{3.912}{1.946} \approx 2.010$.

6 **[Foundational] • Target: Visualizing logarithmic functions**

The graph shows $f(x) = \ln(x)$, the natural logarithm.

a State the domain and the vertical asymptote shown in the graph. Domain: $x > 0$. Vertical asymptote at $x = 0$, where the curve plunges toward $-\infty$ as $x \to 0^+$.

b State the x-intercept shown. x-intercept at $(1, 0)$, since $\ln(1) = 0$.

(a) Domain: $x > 0$. Vertical asymptote at $x = 0$, where the curve plunges toward $-\infty$ as $x \to 0^+$.; (b) x-intercept at $(1, 0)$, since $\ln(1) = 0$.

7 **[Foundational] • Target: Solving exponential equations using logarithms**

Solve $5^x = 40$ for x, to three decimal places.

Taking $\log$ of both sides: $x\log 5 = \log 40$, so $x = \frac{\log 40}{\log 5} \approx \frac{1.602}{0.699} \approx 2.292$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Solving logarithmic equations**

Solve $\log_2(x+3)=4$ for x, and check that the solution keeps the argument positive.

$x+3=2^4=16$, so $x=13$. Check: $x+3=16>0$ ✓. Solution: $x=13$.

9 **[Intermediate]** • **Target: Solving logarithmic equations**

Solve $\log(x)+\log(x-3)=1$ for x, checking both solutions against the domain.

Combine: $\log[x(x-3)]=1$, so $x(x-3)=10$: $x^2-3x-10=0$, $(x-5)(x+2)=0$: $x=5$ or $x=-2$. Since $x>0$ and $x-3>0$ are required (domain of the original logs), $x=-2$ is rejected. Solution: $x=5$.

10 **[Intermediate]** • **Target: Applications: the pH scale and decibels**

The pH of a solution is given by $\text{pH}=-\log[H^+]$, where $[H^+]$ is the hydrogen ion concentration in mol/L. Find the pH of a solution with $[H^+]=3.2\times10^{-5}$ mol/L, to two decimal places.

$\text{pH}=-\log(3.2\times10^{-5})\approx-(-4.495)=4.49$.

11 **[Intermediate]** • **Target: Applications: the pH scale and decibels**

Sound intensity level in decibels is $L=10\log\left(\frac{I}{I_0}\right)$, where I_0 is a reference intensity. If one sound is 1000 times more intense than another, find the difference in their decibel levels.

Difference $=10\log(1000)-10\log(1)=10\log(1000)=10(3)=30$ decibels.

12 **[Intermediate]** • **Target: Transformations of logarithmic functions**

Describe how the graph of $g(x)=\ln(x+2)-3$ is transformed from the parent function $f(x)=\ln(x)$, and state its new vertical asymptote and domain.

$g(x)=\ln(x+2)-3$ shifts $f(x)=\ln(x)$ left 2 units and down 3 units. The vertical asymptote shifts from $x=0$ to $x=-2$, and the domain becomes $x>-2$.

13 **[Intermediate]** • **Target: Transformations of logarithmic functions**

Find the domain of $h(x)=\log(4-x^2)$.

Require $4-x^2>0$: $x^2<4$, so $-2<x<2$. Domain: $(-2,2)$.

14 **[Intermediate]** • **Target: Solving equations that combine exponentials and logarithms**

Solve $e^{2x}-5e^x+6=0$ by substituting $u=e^x$.

Let $u=e^x$: $u^2-5u+6=0$, so $(u-2)(u-3)=0$, giving $u=2$ or $u=3$. Since $u=e^x>0$ for all real x, both are valid: $e^x=2\Rightarrow x=\ln 2$, or $e^x=3\Rightarrow x=\ln 3$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging] • Target: Solving equations that combine exponentials and logarithms • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros**

Solve $\log_2(x) + \log_2(x-2) = 3$ for x, checking the domain.

Combine: $\log_2[x(x-2)] = 3$, so $x(x-2) = 2^3 = 8$: $x^2 - 2x - 8 = 0$, $(x-4)(x+2) = 0$: $x = 4$ or $x = -2$. Domain requires $x > 0$ and $x - 2 > 0$ (i.e. $x > 2$), so $x = -2$ is rejected. Solution: $x = 4$.

16 **[Challenging] • Target: Comparing logarithmic growth to other function families • Cross-Lesson: 1.02 Rational Functions & Asymptotes**

Explain why logarithmic functions grow much more slowly than linear functions for large x, and give a numerical comparison of $\log(x)$ and x at $x = 1000$.

Logarithms compress large inputs: each tenfold increase in x adds only 1 to $\log(x)$, while a linear function adds proportionally to the increase in x itself. At $x = 1000$: $\log(1000) = 3$, while the linear function $y = x$ gives 1000 — the logarithm is vastly smaller, illustrating its extremely slow long-run growth.

17 **[Challenging] • Target: Modeling with logarithms: the Richter scale • Cross-Lesson: 1.03 Rates of Change & Concavity**

Earthquake magnitude on the Richter scale is $M = \log\left(\frac{I}{I_0}\right)$, where I_0 is a reference intensity. An earthquake measures magnitude 7, while another measures magnitude 5. Find how many times more intense the magnitude-7 earthquake is.

$7 - 5 = \log\left(\frac{I_7}{I_0}\right) - \log\left(\frac{I_5}{I_0}\right) = \log\left(\frac{I_7}{I_5}\right) = 2$, so $\frac{I_7}{I_5} = 10^2 = 100$. The magnitude-7 earthquake is 100 times more intense.

18 **[Challenging] • Target: Modeling with logarithms: the Richter scale • Cross-Lesson: 2.01 Exponential Functions & Growth Models**

If an earthquake's intensity is $10{,}000$ times a reference intensity I_0, find its Richter magnitude.

$M = \log(10000) = \log(10^4) = 4$.

19 **[Challenging] • Target: Comparing logarithmic growth to other function families • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle**

Level 1 · FoundationsOn Section I Part B of the AP Precalculus exam, a student allocates 50 minutes to answer 12 questions using a graphing calculator. What is the average time per question available in minutes, rounded to the nearest tenth?

Calculate the ratio of total time to questions: $\frac{50 \text{ min}}{12 \text{ questions}} = \frac{25}{6} \approx 4.167 \text{ min/question}$. Rounded to the nearest tenth, this is 4.2 minutes per question.

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20 [Challenging] • Target: Modeling with logarithms: the Richter scale • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

Find the exact value of $\cos\left(\frac{7\pi}{12}\right)$ and $\sin\left(\frac{7\pi}{12}\right)$ using angle addition identities.

Step 1: Decompose $\frac{7\pi}{12} = \frac{\pi}{3} + \frac{\pi}{4}$: Step 2: Apply cosine and sine sum formulas:

$\cos\left(\frac{7\pi}{12}\right) = \cos\frac{\pi}{3}\cos\frac{\pi}{4} - \sin\frac{\pi}{3}\sin\frac{\pi}{4} = \left(\frac{1}{2}\right)\left(\frac{\sqrt{2}}{2}\right) - \left(\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{2}}{2}\right) = \frac{\sqrt{2}-\sqrt{6}}{4}$. .

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Converting between exponential and logarithmic form

Sound intensity level in decibels is $L = 10\log\left(\frac{I}{I_0}\right)$. If one sound is 10 times more intense than another, find the difference in their decibel levels.

a. 30

b. 20

c. 10 ✓

d. 1

Difference $= 10\log(10) - 10\log(1) = 10(1) = 10$ decibels.

22 [Foundational] • Target: Properties of logarithms

Expand $\log\left(\frac{x}{y}\right)$ using properties of logarithms. (variant 6)

a. $\frac{\log x}{\log y}$

b. $\log(x) \cdot \log(y)$

c. $\log x + \log y$

d. $\log x - \log y$ ✓

Quotient rule: $\log(x/y) = \log x - \log y$.

23 [Foundational] • Target: Visualizing logarithmic functions

Expand $\log\left(\frac{x^2}{y^3}\right)$ using properties of logarithms.

a. $\log(x) \cdot \log(y)$

b. $2\log x - 3\log y$ ✓

c. $2\log x + 3\log y$

d. $\frac{\log x}{\log y}$

Combine power and quotient rules.

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24 [Foundational] • Target: Solving exponential equations using logarithms

Solve $7^x = 100$ for x, expressed using logarithms.

a. $x = \frac{100}{7}$

b. $x = 100 - 7$

c. $x = \log_7(100) = \frac{\ln 100}{\ln 7}$ ✓

d. $x = \log_{100}(7)$

Take $\ln$ of both sides: $x \ln 7 = \ln 100$, so $x = \frac{\ln 100}{\ln 7} \approx 2.367$.

25 [Foundational] • Target: Solving logarithmic equations

Solve: $2^{x+1} = 5$. Express x using logarithms.

a. $x = \frac{\ln 5}{\ln 2} - 1$

b. $x = \frac{\ln 5 - 1}{\ln 2}$

c. $x = \log_2 5 - 1$

d. **Both A and C** ✓

$x + 1 = \log_2 5 \Rightarrow x = \log_2 5 - 1 = \ln 5/\ln 2 - 1$. A, C are equivalent.

26 [Foundational] • Target: Applications: the pH scale and decibels

Prove $a^{\log_a x} = x$ using natural logarithms.

a. Only for $a = e$

b. Not always true

c. **Take** $\ln$**:** $\ln(a^{\log_a x}) = \log_a x \cdot \ln a = \frac{\ln x}{\ln a} \cdot \ln a = \ln x$**. So** $a^{\log_a x} = e^{\ln x} = x$ ✓

d. By induction

Change of base and inverse properties.

27 [Foundational] • Target: Transformations of logarithmic functions

The function $f(x) = -2\cos(\frac{\pi}{3}x) + 5$ has amplitude, period, and midline

a. **2, 6, y=5** ✓

b. 2, 3, y=5

c. –2, 6, y=5

d. 2, 6, y=0

Amplitude |A|=2, period = 2π/(π/3) = 6, midline y = 5.

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28 [Foundational] • Target: Solving equations that combine exponentials and logarithms

A function is one-to-one if:

a. Every output repeats

b. Passes the vertical line test only

c. **Passes the horizontal line test ✓**

d. Is a polynomial

Horizontal line test confirms 1-1.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Comparing logarithmic growth to other function families

The function f is defined by $f(x) = a \cdot b^x + c$, where a, b, and c are constants with $a > 0$, $b > 0$, and $beq1$. In the xy-plane, the horizontal asymptote of the graph of $y = f(x)$ is the line $y = -4$. If the graph passes through the points $(0, -1)$ and $(2, 8)$, what is the value of $f(3)$?

a. **20 ✓**

b. 24

c. 32

d. 44

Step 1: Determine the vertical shift parameter c. For an exponential function in the form $f(x) = a \cdot b^x + c$, the term $a \cdot b^x$ approaches 0 as $x \to -\infty$ (for $b > 1$) or as $x \to \infty$ (for 00), we find $b = 2$. Step 4: Evaluate $f(3)$. The complete function is $f(x) = 3 \cdot 2^x - 4$. Evaluating at $x = 3$:
$f(3) = 3 \cdot 2^3 - 4 = 3(8) - 4 = 24 - 4 = 20$ Misconception Traps: - Choice B (24) results from computing $a \cdot b^3 = 3 \cdot 8 = 24$ but forgetting to subt.

30 [Intermediate] • Target: Modeling with logarithms: the Richter scale

The function $f(x) = a \cdot 2^x + k$ has a horizontal asymptote at $y = 3$ and passes through the point $(0, 7)$, where a and k are real constants. What is the value of $f(2)$?

a. 19 ✓

b. 31

c. 16

d. 11

For an exponential function of the form $f(x) = a \cdot 2^x + k$, as $x \to -\infty$, $2^x \to 0$, which gives the horizontal asymptote $y = k$. Since the horizontal asymptote is given as $y = 3$, we have: $k = 3$ Next, the graph passes through $(0, 7)$, so $f(0) = 7$: $a \cdot 2^0 + 3 = 7$ Since $2^0 = 1$, this simplifies to: $a + 3 = 7 \Rightarrow a = 4$ Thus, the function is $f(x) = 4 \cdot 2^x + 3$. Evaluating at $x = 2$: $f(2) = 4 \cdot 2^2 + 3 = 4 \cdot 4 + 3 = 16 + 3 = 19$ Common errors: - Se.

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31 [Intermediate] • Target: Converting between exponential and logarithmic form

What is the horizontal asymptote of the graph of the function $f(x) = 3 \cdot 2^{x+1} - 5$?

a. $y = -5$ ✓

b. $y = 0$

c. $y = 3$

d. $x = -1$

An exponential function in the standard form $f(x) = a \cdot b^{x-h} + k$ (with base $b > 1$) has a horizontal asymptote determined by its vertical shift, $y = k$. As $x \to -\infty$, the exponential power $2^{x+1} \to 0$. Substituting this into the function yields: $\lim_{x \to -\infty} f(x) = 3(0) - 5 = -5$ Thus, the horizontal asymptote is the horizontal line $y = -5$. Common Misconceptions: - $y = 0$: Assumes the asymptote is always the x-axis, forgetting the vertical shift by -5..

32 [Intermediate] • Target: Properties of logarithms

The graph of the exponential function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$. Determine the value of a and the value of b.

a. **a = 4, b = 3** ✓

b. a = 3, b = 4

c. a = 6, b = 2

d. a = 2, b = 6

Let the function be $f(x) = a \cdot b^x$. We are given that the graph passes through $(1, 12)$ and $(3, 108)$. Substituting these points into the function equation, we get two equations: 1) $a \cdot b^1 = 12 \Rightarrow ab = 12$ 2) $a \cdot b^3 = 108$ From equation (1), we can express a in terms of b: $a = \frac{12}{b}$. Substitute this expression for a into equation (2): $\left(\frac{12}{b}\right) \cdot b^3 = 108$ $12 \cdot b^2 = 108$ $b^2 = \frac{108}{12}$ $b^2 = 9$ Since b must be positive for an exponen.

33 [Intermediate] • Target: Visualizing logarithmic functions

The graph of the exponential function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$. Determine the value of a.

a. 2

b. 3

c. **4** ✓

d. 6

Since the graph of $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$, we can set up a system of two equations: 1. $a \cdot b^1 = 12 \Rightarrow ab = 12$ 2. $a \cdot b^3 = 108$ From equation (1), we can express a in terms of b: $a = \frac{12}{b}$. Substitute this into equation (2): $\left(\frac{12}{b}\right) \cdot b^3 = 108$ $12b^2 = 108$ $b^2 = \frac{108}{12}$ $b^2 = 9$ Since b is the base of an exponential function, $b > 0$. Thus, $b = 3$. Now, substitute the value of b back into equ.

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34 [Intermediate] • Target: Solving exponential equations using logarithms

What is the value of the exponential function $f(x) = 2^x$ when $x = 3$?

a. 6

b. **8** ✓

c. 9

d. 5

The exponential function is given by $f(x) = 2^x$. To find the value of the function when $x = 3$, we substitute 3 for x into the function's formula: $f(3) = 2^3$ $2^3 = 2 \times 2 \times 2 = 8$ Therefore, the value of the function $f(x) = 2^x$ when $x = 3$ is 8.

35 [Intermediate] • Target: Solving logarithmic equations

What is the value of y when $x = 3$ for the function $f(x) = 2^{x+1}$?

a. 4

b. **8** ✓

c. 6

d. 16

To find the value of y when $x = 3$ for the function $f(x) = 2^{x+1}$, we substitute $x = 3$ into the function: $f(3) = 2^{3+1}$ $f(3) = 2^4$ $f(3) = 16$ Therefore, when $x = 3$, $y = 16$.

36 [Intermediate] • Target: Applications: the pH scale and decibels

Write using natural log: $\log_b x =$

a. $\frac{\ln b}{\ln x}$

b. $\ln x - \ln b$

c. $\ln x \cdot \ln b$

d. $\frac{\ln x}{\ln b}$ ✓

Applying Exponential & Logarithmic Functions principles, Change of base formula. Therefore, the verified correct option is $\frac{\ln x}{\ln b}$.

37 [Intermediate] • Target: Transformations of logarithmic functions

Convert to exponential form: $\log_5 25 = 2$.

a. $5^{25} = 2$

b. $2^5 = 25$

c. $5^2 = 25$ ✓

d. $25^2 = 5$

Applying Exponential & Logarithmic Functions principles, Log to exponential. Therefore, the verified correct option is $5^2 = 25$.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Solving equations that combine exponentials and logarithms • Cross-Lesson: 2.01 Exponential Functions & Growth Models

Solve: $3^x = 2^{x+1}$. Use logarithms.

a. $x = \frac{\ln 2}{\ln 3}$

b. $x = 1$

c. $x = \frac{\ln 3}{\ln 2}$

d. $x = \frac{\ln 2}{\ln 3 - \ln 2}$ ✓

$x\ln 3 = (x+1)\ln 2 \Rightarrow x(\ln 3 - \ln 2) = \ln 2 \Rightarrow x = \frac{\ln 2}{\ln 3 - \ln 2}$.

39 [Challenging] • Target: Comparing logarithmic growth to other function families • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

Show that $e^{i\pi} + 1 = 0$ using Euler's formula.

a. $e^{i\pi} = i$

b. $e^{i\pi} = \cos\pi + i\sin\pi = -1 \Rightarrow e^{i\pi} + 1 = 0$ ✓

c. $e^{i\pi} = 1$

d. Euler's formula is $e^{ix} = \sin x + i\cos x$

Euler: $e^{i\theta} = \cos\theta + i\sin\theta$. At $\theta = \pi$: $-1 + 0i = -1$.

40 [Challenging] • Target: Modeling with logarithms: the Richter scale • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

Write $5^4 = 625$ in logarithmic form.

a. $\log_{625}(5) = 4$

b. $\log_5(4) = 625$

c. $\log_4(5) = 625$

d. $\log_5(625) = 4$ ✓

$b^x = y \Leftrightarrow \log_b(y) = x$, so $\log_5(625) = 4$.

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41 [Challenging] • Target: Converting between exponential and logarithmic form • Cross-Lesson: 4.01 Inverse & Composite Functions

The function $f(x) = x - e^x$ has exactly one critical point. Find it.

a. $x = e$

b. $x = 1$

c. $x = -1$

d. $x = 0$ ✓

$f'(x) = 1 - e^x = 0 \Rightarrow e^x = 1 \Rightarrow x = 0$. Maximum.

42 [Challenging] • Target: Properties of logarithms • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

Convert to logarithmic form: $3^4 = 81$.

a. $\log_4 81 = 3$

b. $\log_3 81 = 4$ ✓

c. $\log_{81} 3 = 4$

d. $\log_3 4 = 81$

Exponential to log: $a^b = c \Leftrightarrow \log_a c = b$.

43 [Challenging] • Target: Visualizing logarithmic functions • Cross-Lesson: 4.03 Parametric Equations & Modeling

Express $\log_3 7$ using the change of base formula with natural log.

a. $\frac{\ln 7}{\ln 3}$ ✓

b. $7 \ln 3$

c. $\frac{\ln 3}{\ln 7}$

d. $\ln 7 - \ln 3$

Change of base: $\log_b x = \frac{\ln x}{\ln b}$.

44 [Challenging] • Target: Solving exponential equations using logarithms • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

Which function is the inverse of $f(x) = e^x$?

a. $g(x) = x^e$

b. $g(x) = \ln x$ ✓

c. $g(x) = 10^x$

d. $g(x) = \log x$

Natural log is inverse of natural exponential.

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45 [Challenging] • Target: Solving logarithmic equations • Cross-Lesson: 1.02 Rational Functions & Asymptotes

The function $f(x) = e^x - x - 1$ satisfies $f(x) \geq 0$. Prove it.

a. Only for $x \geq 0$

b. $f(0) = 0$, $f'(x) = e^x - 1 = 0$ **at** $x = 0$ **(minimum).** $f''(0) = e^0 = 1 > 0$**: global min at** $x = 0$ **with value 0.** ✓

c. Not true

d. By AM-GM

$e^x \geq 1 + x$ for all x (known inequality).

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